

ANALYTICS | STRATEGIC INSIGHTS

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

EXECUTIVE SUMMARY

\$233K

Total revenue analyzed across
3,900 transactions

80%

Of customers already classified
as “**Loyal**” repeat buyers

27%

Of customers hold an **active**
subscription

48%

Average discount rate on top 5
discount-dependent SKUs

Key Narrative

- Male customers generate **2.1x** the revenue of female customers a segmentation gap worth investigating further.
- **Non-subscribers drive 73% of total revenue**, but spend at nearly the same average ticket as subscribers (\$59.87 vs. \$59.49)
subscription is not yet a spend multiplier.
- Discount dependency is concentrated in a small set of products (Hats, Sneakers, Coats), suggesting these categories may be **over-discounted** relative to demand.
- Express shipping customers spend **3.4%** more per order than Standard shipping customers, hinting at a willingness-to-pay premium.

INTRODUCTION

Background

Retail leadership needs a clearer view of who is buying, what drives repeat behavior, and where margin is being given away through discounting. This project consolidates transactional, demographic, and behavioral data into a single analytical view to support that decision-making.

Scope of Analysis

- **Customer demographics** — age, gender, location, subscription status
- **Purchase details** — item, category, amount, season, size, color
- **Shopping behavior** — discounts, promo codes, frequency, ratings, shipping

Dataset at Glance

3850+
Transaction

17+
Data Fields

25+
Product Types

50+
Customer Locations

OBJECTIVE

Spending Patterns

Where is revenue concentrated across gender, age, category, and shipping method?

Customer Segments

Who are our New, Returning, and Loyal customers, and how do they behave differently?

Product Preferences

Which products win on rating and volume, and which rely too heavily on discounting?

Subscription Behavior

Does a subscription actually change spending, and how can it be made more valuable?

METHODOLOGY

Phase 1: Data Preparation

used Python to clean the data

Data Loading – Imported and read the dataset using pandas.

Initial Exploration – Used `df.info()` to check structure and `.describe()` for summary statistics.

Missing Data Handling – Checked for null values; found 37 missing values in the Review Rating column; imputed them using the median rating of each product category.

Column Standardization – Renamed columns to snake_case for readability and documentation.

Feature Engineering – Created an `age_group` column by binning customer ages, and a `purchase_frequency_days` column derived from purchase data.

Data Consistency Check – Verified whether `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

Database Integration – Connected the Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

METHODOLOGY

Phase 2: Business Analysis (SQL)

Ran 10 structured business queries

Revenue by Gender – Compared total revenue generated by male vs. female customers.

High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

Top 5 Products by Rating – Found the highest-rated products by average review score.

Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

Discount-Dependent Products – Identified the 5 products with the highest percentage of discounted purchases.

Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

Top 3 Products per Category – Listed the most purchased products within each product category.

Repeat Buyers & Subscriptions – Checked whether customers with more than 5 purchases were more likely to subscribe.

Revenue by Age Group – Calculated the total revenue contribution of each age group.

METHODOLOGY

Phase 3: Visualization

Used Power BI to build dashboard

Dashboard Design – Built an interactive Power BI dashboard to present insights visually for business stakeholders.

KPI Cards – Surfaced headline metrics: number of customers, average purchase amount, and average review rating.

Subscription Breakdown – Added a donut chart showing % of customers by subscription status.

Category Performance Views – Created bar charts for sales and revenue by category.

Age Group Views – Added bar charts for revenue and sales by age group.

Interactive Filtering – Enabled slicers for subscription status, gender, category, and shipping type so stakeholders can explore the data themselves.

METHODOLOGY

Phase 4: Insight Synthesis & Reporting

Insight Consolidation – Combined results from all 10 SQL queries and the Power BI dashboard into one analytical narrative.

Root-Cause Analysis – Examined patterns together (e.g., subscriber spend, discount reliance, gender split) to identify why, not just what.

Solution Ideation – Mapped each root cause to a candidate action (e.g., low subscriber premium → redesign perks; discount dependency → policy audit).

Prioritization – Ranked actions by impact vs. effort to separate quick wins from longer-term initiatives.

Reporting – Structured findings into a report and deck following: **overview → data → insights → causes → recommendations → roadmap**.

PERFORMANCE SNAPSHOT

\$59.76

Average Purchase Amount

3.75/5

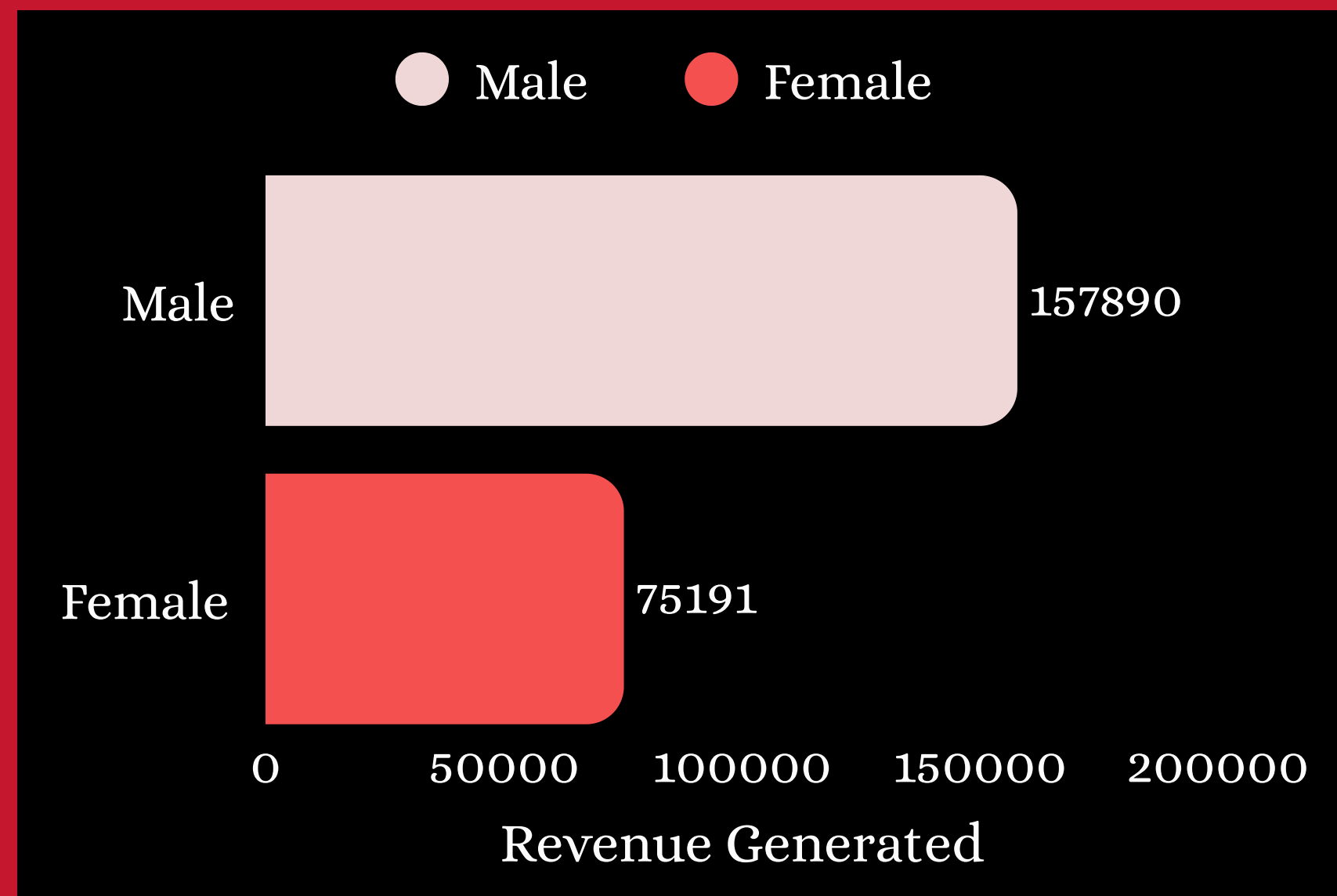
Average Review Rating

27%

Active Subscribers

KEY INSIGHTS

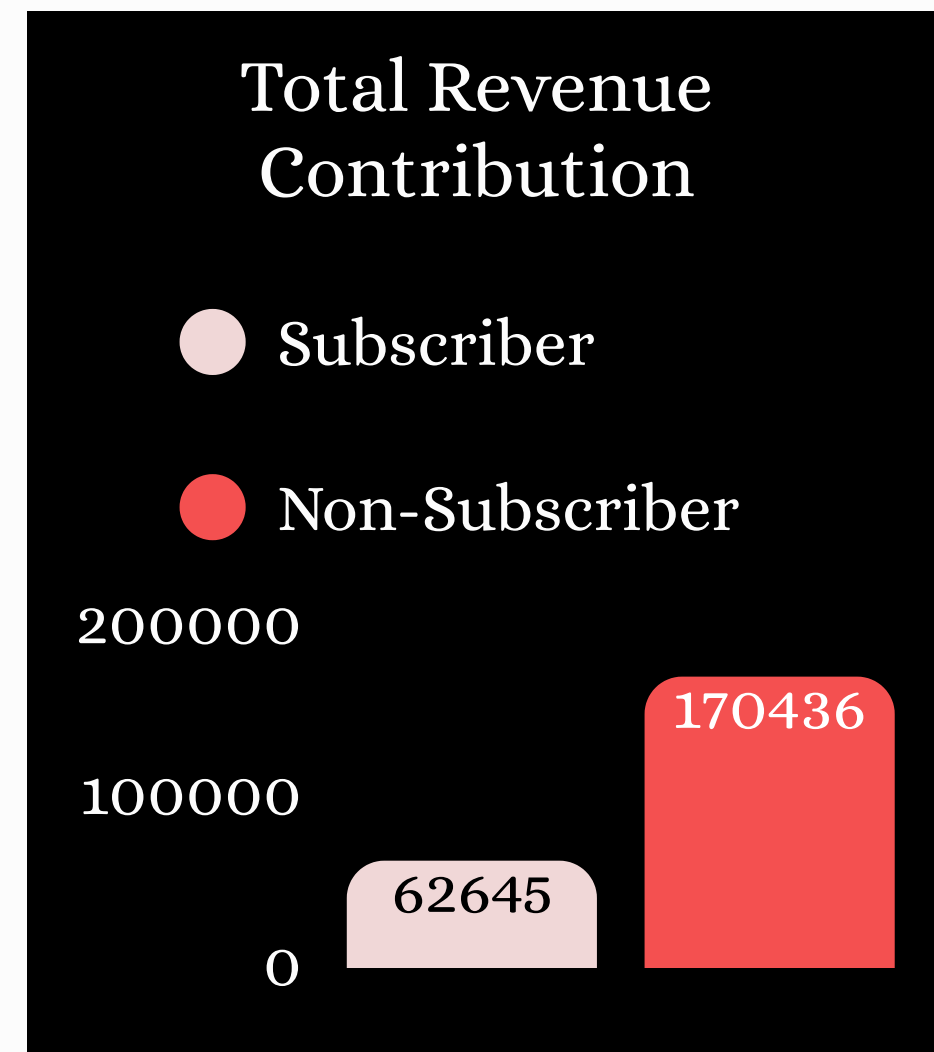
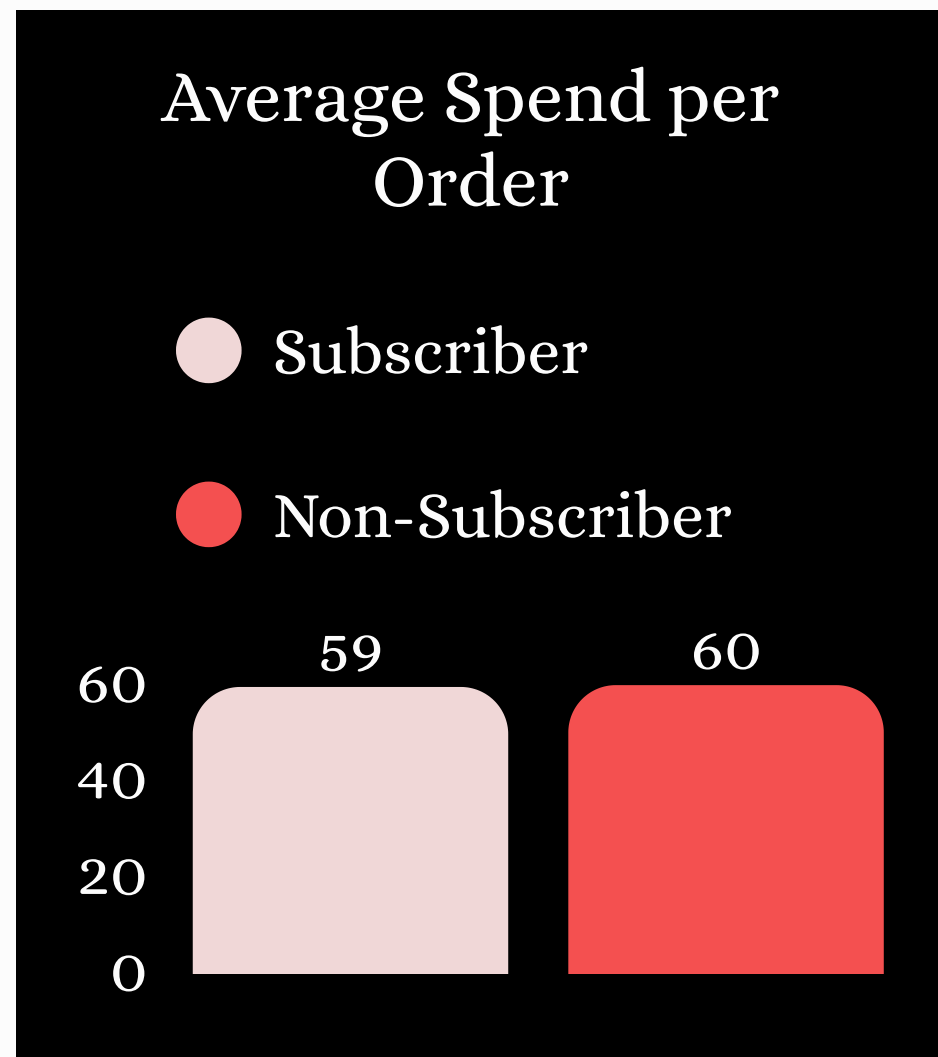
Male customers generate over 2x female revenue



- Male customers account for **~68%** of total revenue in this dataset, despite comparable purchase counts across genders.
 - This may reflect category mix (e.g. higher-ticket outerwear/footwear) rather than higher per-customer loyalty.
- Recommend a follow-up cut by category × gender to confirm whether this is a basket-size or product-mix effect

KEY INSIGHTS

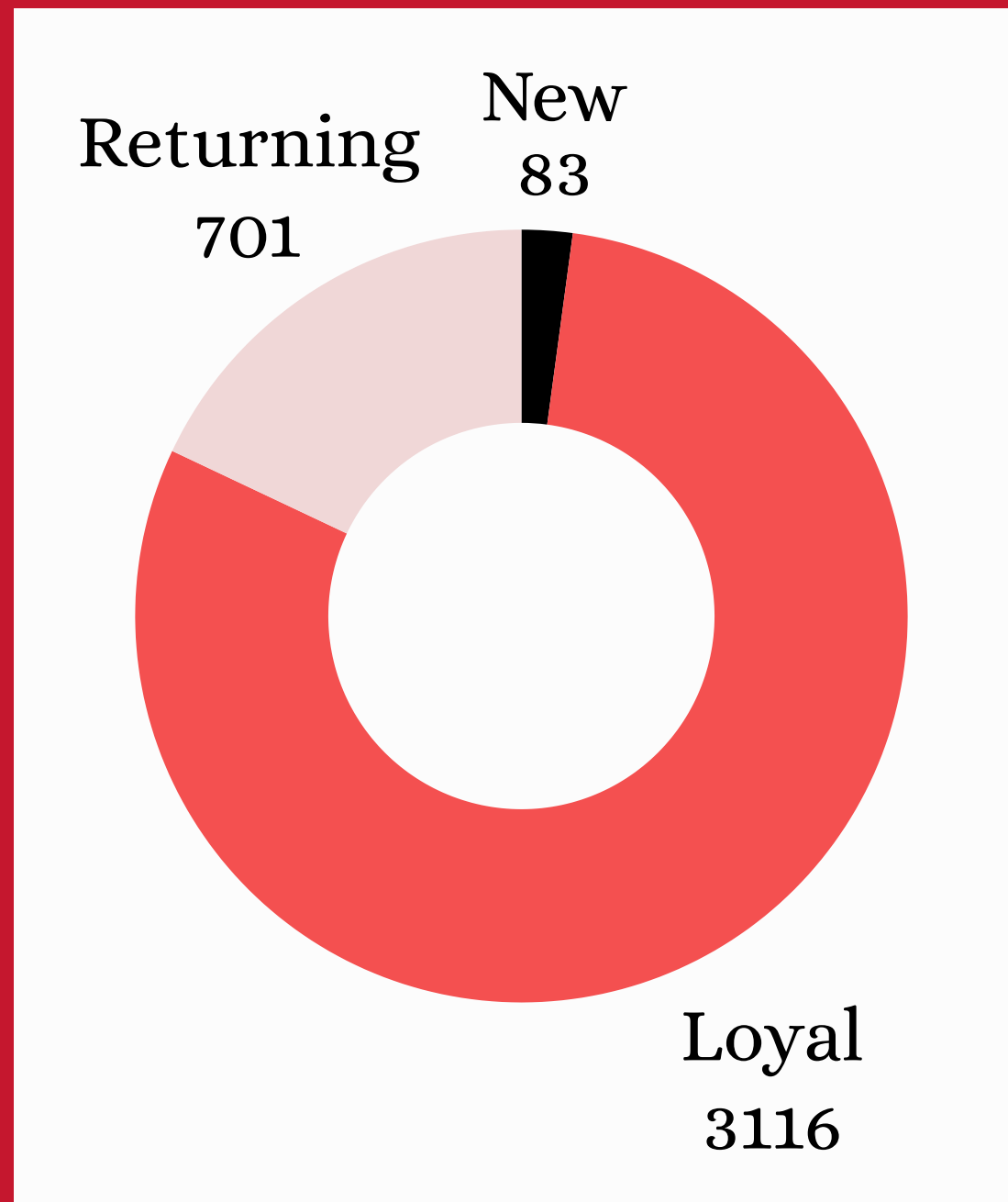
Subscription drives volume, not yet a spending premium



- 1,053 subscribers spend almost identically to 2,847 non-subscribers (\$59.49 vs. \$59.87) — subscription is a retention tool today, not an upsell engine.
- Non-subscribers still generate 73% of total revenue, so broad-based improvements matter as much as subscriber-only perks.
- Repeat buyers (>5 purchases) split 2,518 non-subscribers to 958 subscribers — loyalty is happening outside the subscription program.
- Opportunity: redesign subscription benefits to convert already-loyal, high-frequency shoppers.

KEY INSIGHTS

80% of customers are already “Loyal” - protect this base



LOYAL (80%)-
High purchase frequency — core revenue base to protect and reward

RETURNING (18%)-
Multiple purchases, not yet habitual — prime nurture target

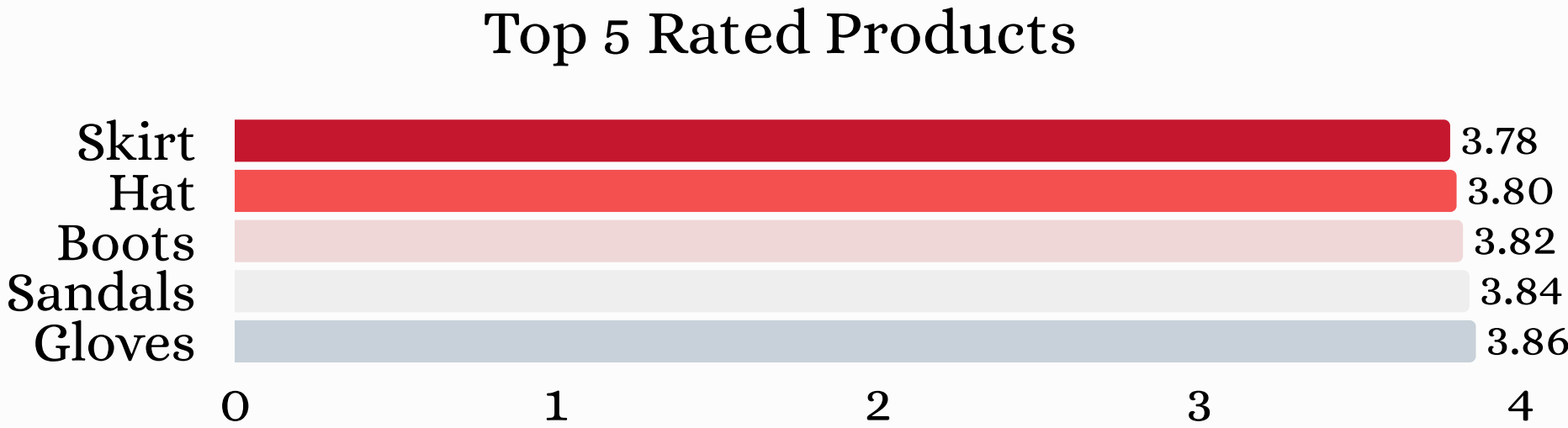
NEW (2%)-
First-time buyers — small pool, needs acquisition investment

KEY INSIGHTS

Clear category leaders — but ratings cluster tightly

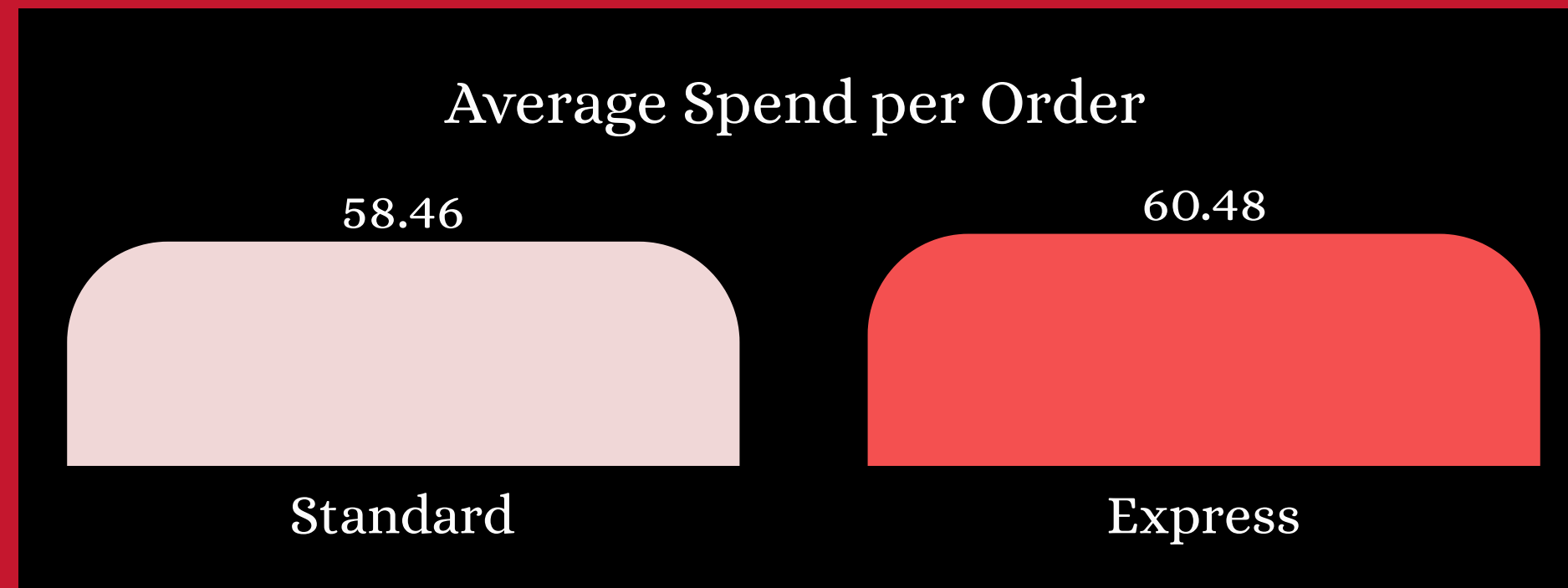
Top Selling item per category		
Category	#1 Item	Orders
Clothing	Blouse	171
Accesories	Jewellery	171
Outerwear	Jacket	163
Footwear	Sandals	160

- Category leaders (Blouse, Jewelry, Jacket, Sandals) each move 160+ orders — strong, evenly-spread demand rather than one runaway hit.
- Top-rated products cluster narrowly between 3.78–3.86 — no product is a clear quality standout, and none is a clear laggard either.
- This flat rating distribution suggests customer satisfaction is consistent across the catalog, so differentiation should come from merchandising and pricing, not just product quality fixes.

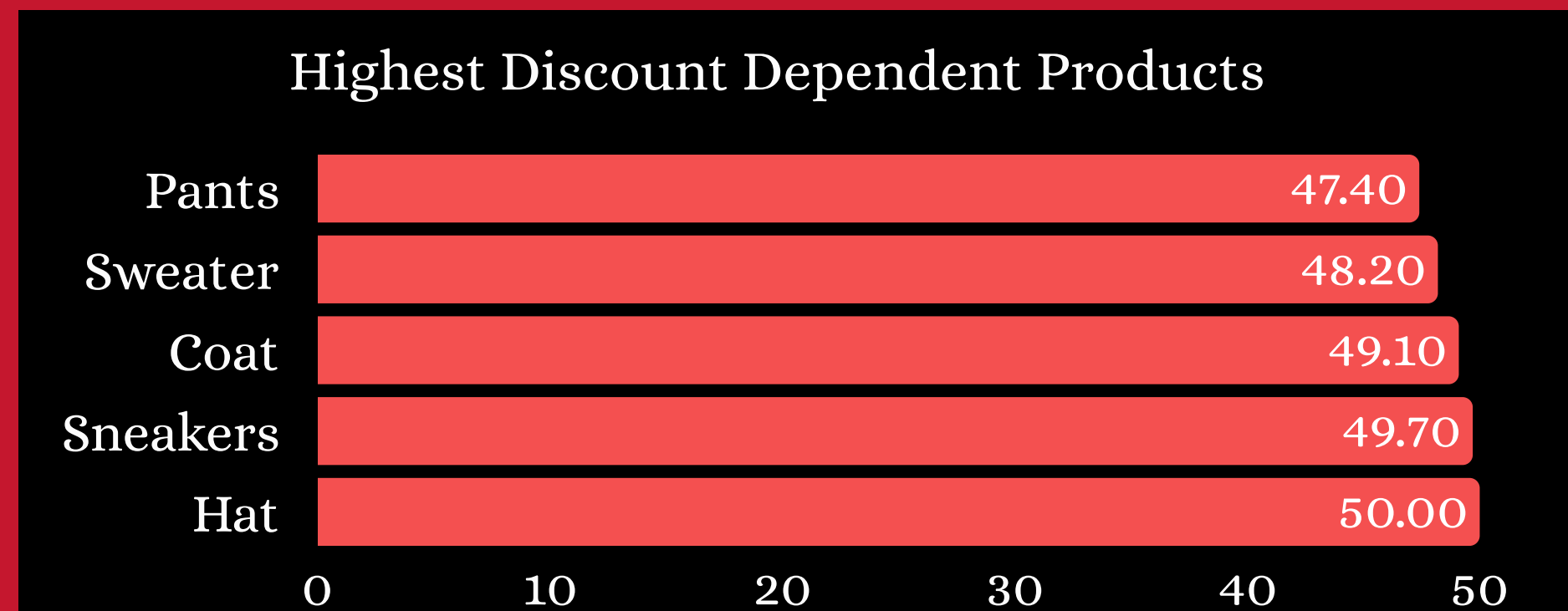


KEY INSIGHTS

Discounting is concentrated and shipping signals a premium



Express-shipping customers spend 3.4% more per order than Standard — a small but consistent signal of price insensitivity worth testing with premium bundles.



Roughly half of every sale in these 5 products involves a discount — far above the catalog average.

ROOT CAUSE ANALYSIS

Observed Issue	Likely Root Cause	Impact
Subscription revenue share is low (27%)	Subscriber and non-subscriber spend per order are nearly identical, so there is little visible incentive to convert or stay subscribed.	High
Heavy reliance on discounts in 5 core SKUs	These categories (Hats, Sneakers, Coats, Sweaters, Pants) may be mispriced at full price relative to perceived value, or discounting has become habitual.	High
Gender revenue gap (Male 2.1x Female)	Likely driven by category mix and average basket size rather than customer count — needs a category × gender cross-tab to confirm.	Medium
New-customer segment is very small (2%)	Acquisition funnel and first-purchase incentives may be underpowered relative to retention-focused efforts.	Medium

PROPOSED CHANGES

Redesign subscription value

Add spend-based perks (free express shipping, early access) so subscribing measurably increases basket size, not just frequency.

Launch a loyalty program

Formal tiering for the 3,116 “Loyal” customers to defend this base against competitor poaching.

Audit discount policy

Review margin impact on Hats, Sneakers, Coats, Sweaters, and Pants; test smaller, targeted discounts vs. blanket 50% cuts.

Elevate top-rated products

Feature Gloves, Sandals, Boots, Hat, and Skirt in campaigns and homepage placement to convert quality into visibility.

Test a shipping-tier upsell

Package Express shipping with mid-to-high basket sizes, leveraging the observed willingness to pay more.

ROADMAP

PHASE 1

Quick Wins

0-2 Months

- Feature top-rated products in campaigns
- Cap discount depth on top 5 dependent SKUs

PHASE 2

Program & Design

2-4 Months

- Design tiered loyalty program
- Redefine subscription perks & pricing

PHASE 3

Scale & Test

4-8 Months

- Pilot shipping-tier upsell bundles
- Launch category × gender deep-dive campaign

CONCLUSION

This customer base is loyal, satisfied, and spending consistently, the next stage of growth comes from making the subscription program worth more, tightening discount discipline, and pointing marketing at the segments and products already proven to perform.

Summary of findings

- 80% of customers are already loyal repeat buyers — a defensible core to protect
- Subscribers and non-subscribers spend almost identically — subscription needs redesign
- Discounting is concentrated in 5 products at ~48–50% — a margin risk worth auditing
- Express shipping customers pay more — an untapped premium signal

Suggested Next Steps

- Validate category × gender revenue drivers with a deeper cross-tab
- Run a subscription-perk pilot with a control group over one quarter
- Set discount guardrails for the top 5 dependent products
- Revisit this dashboard quarterly to track KPI movement

To explore all the data analysis click on the link below

[LINK](#)

THANK YOU